Hotel Loyalty Points Mall

Software Design Report

CS6451: Advanced Software Design • Autumn 2026–2027
University of Limerick

Team-Based Project

Lecturer: J. J. Collins
Submission: Week 8

Team Members

Fude Ding	26046091
Kewei Wu	26141825



Contents

Hotel Loyalty Points Mall 4

Project Report . . . 4

1. Business Scenario . . . 4

Actual running screenshots . . . 4

Points Mall . . . 4

My Points . . . 5

Add Product . . . 5

Points Adjustment . . . 6

2. Executive Summary . . . 6

Architecture diagrams . . . 7

3. Scope and Use Cases . . . 9

Implementation status at a glance . . . 10

UC1 - Login and account information . . . 10

UC2 - Points transaction details . . . 10

UC3 - Redeem products . . . 10

UC4 - Manage points rules . . . 11

UC5 - Manually add points . . . 11

UC6 - Manage products . . . 11

4. Functional Architecture . . . 12

Architectural patterns . . . 12

Main packages . . . 13

5. Design Patterns . . . 13

5.1 Layered architecture . . . 13

5.2 Strategy pattern for payment . . . 13

5.3 State pattern for orders . . . 13

5.4 Message-driven processing . . . 14

5.5 Optimistic concurrency . . . 14

5.6 Chain of Responsibility for post-login checks . . . 15

5.7 Template Method in login handlers . . . 15

5.8 Observer / publish-subscribe event handling . . . 15

5.9 Factory and Data Mapper patterns . . . 16

5.10 MVC and dependency injection . . . 16

5.11 Repository-level design principles . . . 17

6. Domain Model . . . 18

7. Database Design . . . 19

8. API and Page Endpoints . . . 20

Authentication . . . 20

Guest pages and APIs . . . 20

Manager pages and APIs . . . 20

9. Security and Reliability . . . 21

Authentication and authorization . . . 21

Data and transaction safety . . . 21

Asynchronous reliability . . . 21

Current limitations . . . 21

10. User Interface . . . 21

11. Demo Data and Running the Application . . . 22
Prerequisites . . . 22
Start the application . . . 22
Build and test . . . 22
Demonstration credentials . . . 22
Useful local tools . . . 22
12. Testing Status . . . 22
Testing approach . . . 23
CI/CD . . . 23
13. UML and Supporting Documents . . . 23
14. Implementation Gaps and Recommended Next Steps . . . 24
Priority 1: **[Key gap]** Connect points rules to earning . . . 24
Priority 2: **[Key gap]** Add operational queue management . . . 24
Priority 3: **[Partially complete]** Make product availability explicit . . . 24
Priority 4: **[Partially complete]** Complete order lifecycle parity . . . 24
Test coverage status . . . 24
15. Conclusion . . . 24
16. Technology Stack and Technical References . . . 24
Technology and theory references . . . 25
17. Assignment Deliverables and Traceability Matrix . . . 25
18. Software Lifecycle Discussion . . . 26
19. Project Plan Summary . . . 26
Final project plan . . . 27
20. Quality Attributes and Quality Tactics . . . 27
21. Transparency and Traceability . . . 27
7.1 Package and class counts . . . 28
7.2 Class authorship and lines of code . . . 29
7.3 Team contribution . . . 30
22. Critique of Design and Implementation . . . 31
Utility of diagrams . . . 31
23. Reflection . . . 31
Fordring . . . 31
Franklin . . . 31
24. Generative AI Use . . . 31
25. References and Submission Checklist . . . 32
Formal reference list . . . 32
Final export checklist . . . 32

Hotel Loyalty Points Mall

Project Report

A hotel loyalty points management and redemption system built with Spring Boot, JSP, MyBatis, H2, Spring Security, and embedded Artemis JMS.

Project: CS6451 Advanced Software Design **Application:** `hotel-loyalty` **Version:** `0.0.1-SNAPSHOT` **Java:** 21 **Build:** Maven **Last reviewed:** 4 November 2026

---

1. Business Scenario

Hotel guests earn and spend loyalty points. The hotel publishes redeemable products in a points mall, such as benefits or hotel-related rewards. A guest can pay with points alone, or with a mix of points and another payment method (currently cash or voucher).

Managers maintain the product catalogue and the rules that describe how points are earned. They can also manually award points to a guest. Manual awards go into a queue, so temporary processing failures do not lose the request.

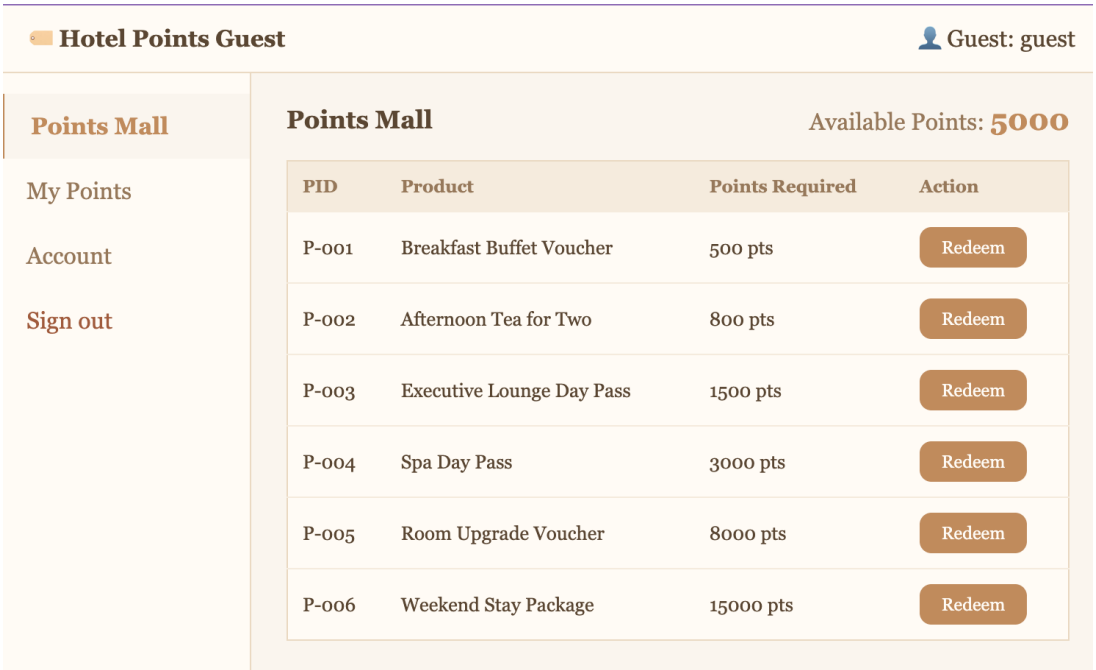
The main business rules are:

- A guest must be logged in and can only access their own resources.
- A manager-only operation must not be available to a guest.
- A redemption cannot use more points than the guest owns.
- A redemption cannot use an unavailable product.
- Remaining stock must never go negative.
- A product with active orders cannot be deleted.
- Every redemption and manual award creates a signed points transaction.

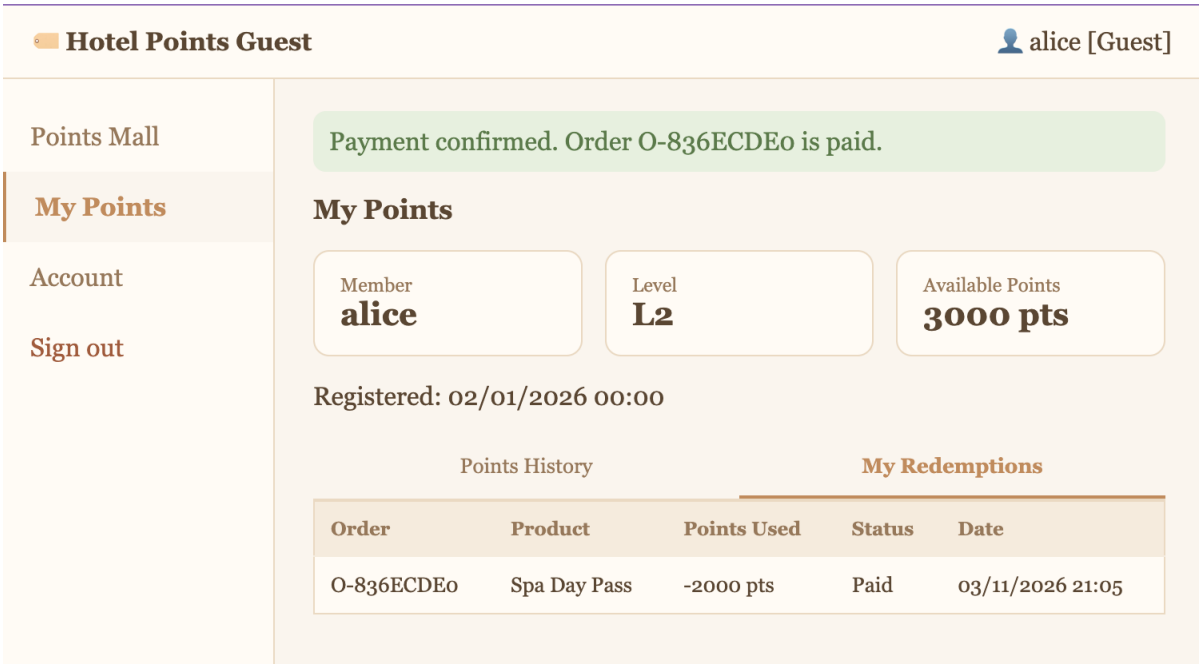
Actual running screenshots

The following screenshots show the implemented application running, rather than only the design-time HTML prototypes:

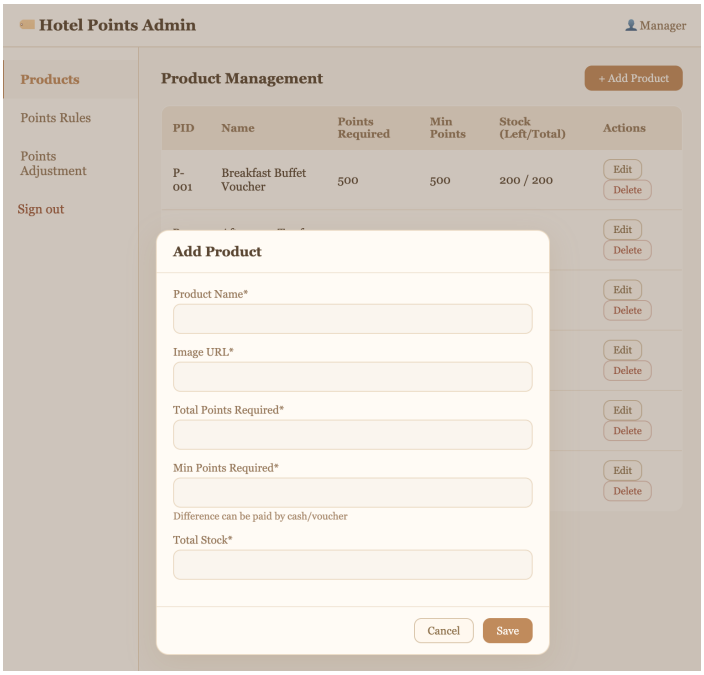
Points Mall



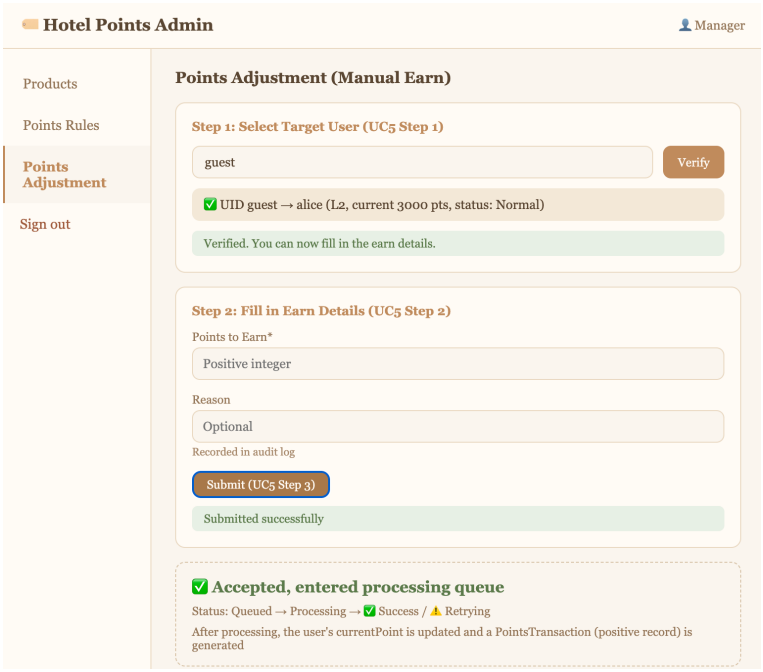
My Points



Add Product



Points Adjustment



---

2. Executive Summary

Hotel Loyalty is a web application for managing a hotel's guest loyalty points and points-mall products. It has two authenticated user roles:

- **Guest**: logs in, views the points mall, checks the current balance, picks a product, chooses a payment method, and redeems the product.
- **Manager**: logs in through the manager portal, manages points-mall products, manages points rules, and manually adds points to a guest account.

The current implementation uses server-rendered JSP pages for the user interface and REST endpoints for browser-side administration and guest operations. Business logic lives in service classes, database access is isolated in MyBatis mappers, and Spring Security controls role-based access.

The implementation also shows several software-design techniques:

1. **Layered architecture**: controller, service, mapper, and persistence layers.

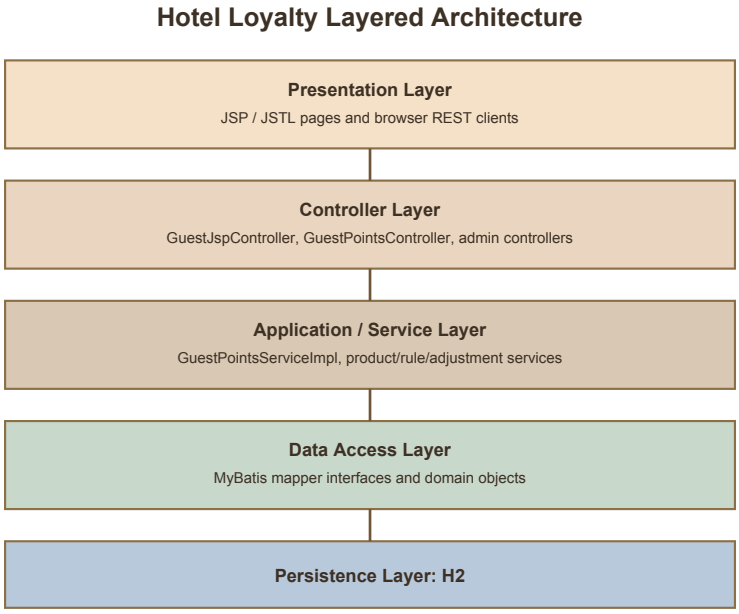
2. **Strategy pattern**: payment validation is selected through `PaymentStrategyFactory`.

3. **State pattern**: order lifecycle states are represented by the classes in the `order` package.

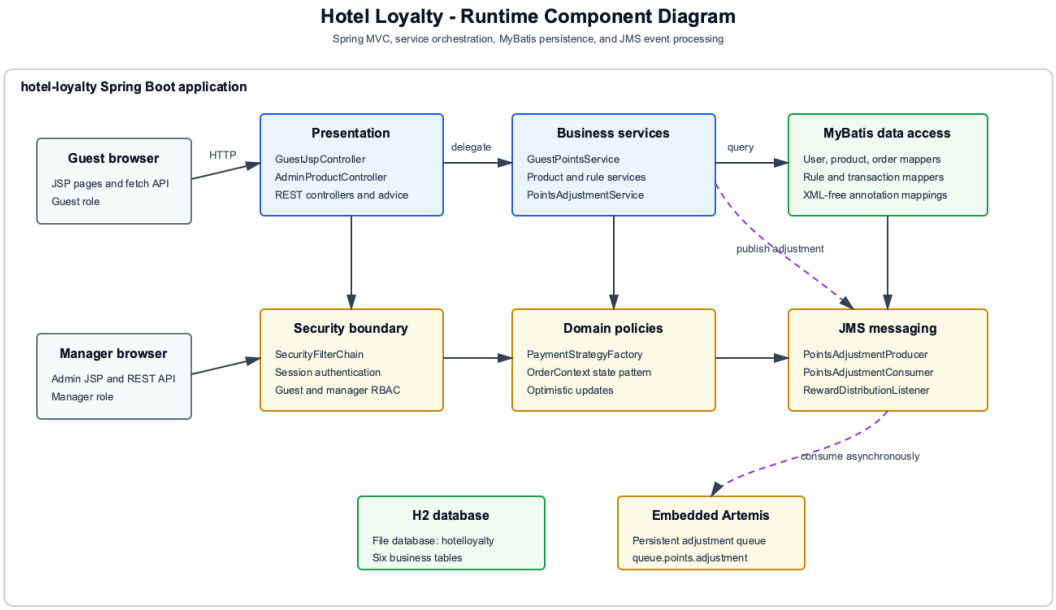
4. **Message-driven processing**: manual point adjustments are sent to an embedded Artemis queue and processed asynchronously.

5. **Optimistic concurrency control**: point balances and stock are updated with compare-and-update operations.

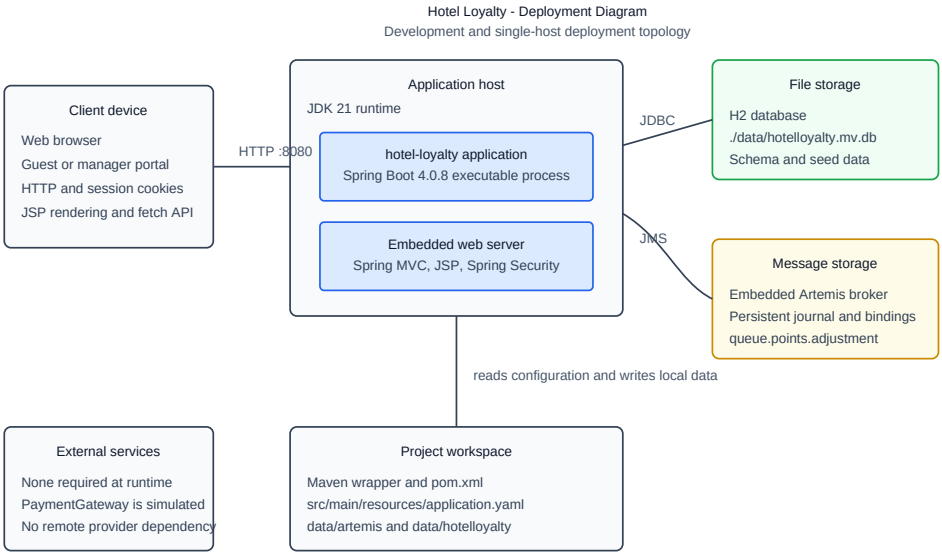
Architecture diagrams



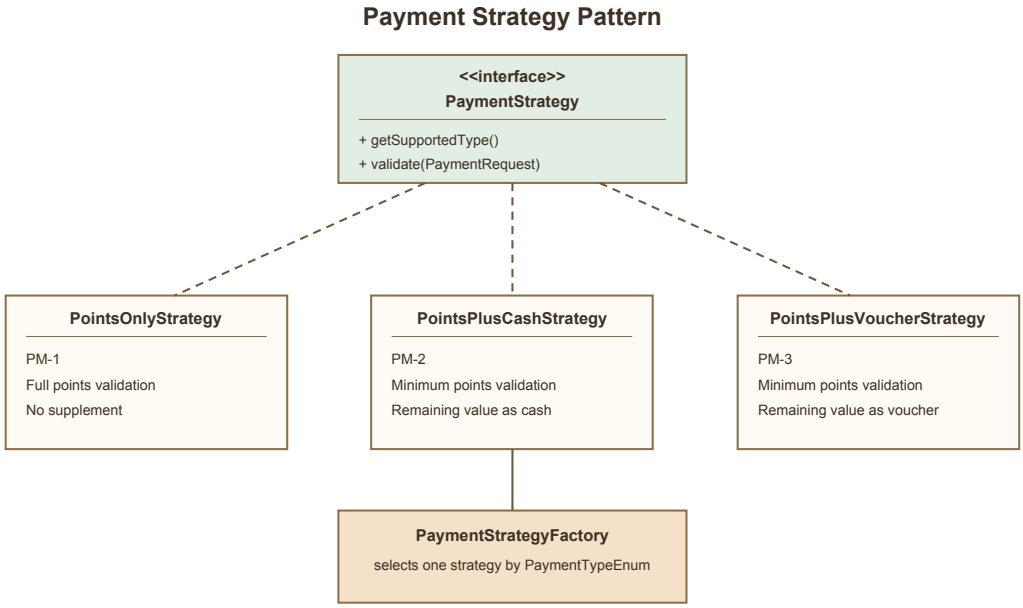
The design-time package boundaries are shown in `package_diagram.svg`. The runtime component collaboration is shown below.



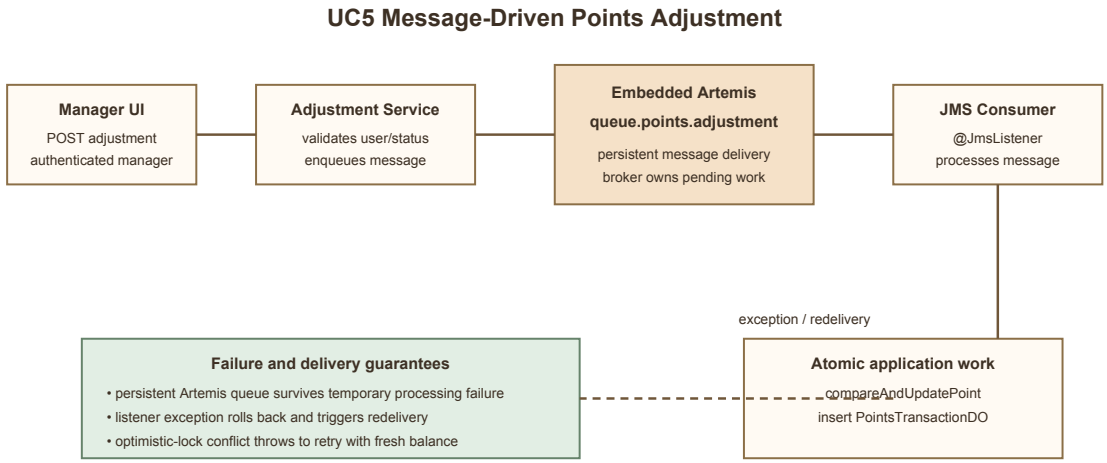
The single-host deployment topology is shown below.



The payment design is a **Strategy pattern**, shown below. It is not a Chain of Responsibility: `PaymentStrategyFactory` picks exactly one strategy from `PaymentTypeEnum`, and strategies do not pass an unhandled request to one another. The controller and service call the chosen strategy directly.



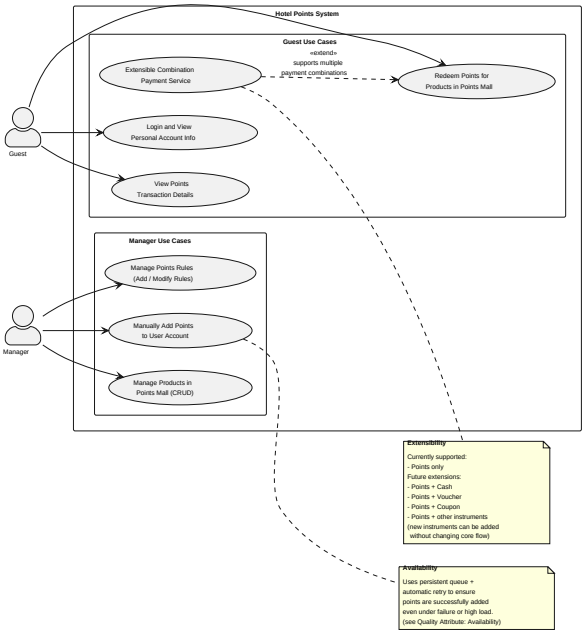
The asynchronous points-adjustment flow and its reliability controls are shown below. The manager request is validated and placed on the persistent `queue.points.adjustment` Artemis queue. The listener updates the user balance and writes the transaction. An exception rolls back processing and causes redelivery; the optimistic compare-and-update also turns concurrent balance conflicts into retryable failures.



This report describes the implemented system, not the original conceptual model. Where a use case or UML artifact describes planned behavior that is not yet connected to the running code, it is explicitly marked as a gap.

3. Scope and Use Cases

The use-case catalogue has six primary use cases.



ID	Use case	Current implementation
UC1	Guest login and personal account information	**[Implemented]** Login, session identity, user ID, name, role, status, level, balance, registration time, and account page are implemented
UC2	View points transaction details	**[Implemented]** My Points shows signed history, type, related order, timestamp, and redemption orders
UC3	Redeem points for products in the points mall	**[Implemented within simulated-payment scope]** Points, simulated cash, and simulated voucher flows validate and persist orders
UC4	Manager manages points rules	**[Partially implemented / key gap]** Rule CRUD works; **hotel-order points calculation and award processing are not connected**

ID	Use case	Current implementation
UC5	Manager manually adds points	**[Core implemented / partially complete]** Persistent queue, consumer, retry, balance update, and EARN transaction work; **queue status, dead-letter handling, request status, and reason persistence are not implemented**
UC6	Manager manages products	**[Core implemented / partially complete]** Product CRUD and validation work; **guest-facing inventory availability needs better presentation**

Implementation status at a glance

These are the **key incomplete or not-yet-implemented features** and should not be described as fully delivered:

- **UC1:** No known missing item within the current use-case scope. The account page shows every field currently stored for the user: ID, name, role, status, level, current points, and registration time.
- **UC3:** No missing item within the agreed prototype scope. Cash and voucher are intentionally simulated; real provider integration is explicitly out of scope.
- **UC4 — Rule execution:** managers can maintain rules, but no hotel-order earning event picks a rule and calculates/awards points.
- **UC5 — Operational queue management:** there is no user-facing queued/processed/retrying/failed status, durable request ID, dead-letter administration, or persisted adjustment reason.
- **UC6 — Availability presentation:** the backend filters available products, but the guest-facing product view does not show a complete stock/availability state for every product.
- **Order lifecycle orchestration:** the redemption service uses the order state classes for the pay transition, but cancellation, fulfilment, and shipment workflows are not implemented.

UC1 - Login and account information

The application has separate login pages:

- `/login` for guests
- `/admin/login` for managers

Authentication uses Spring Security sessions and BCrypt passwords. The guest points page shows the guest name, level, balance, registration time, and redemption records. The account page shows all stored account fields: user ID, name, role, status, membership level, current points, and registration time. **UC1 is complete for the current data model and viewing scope.**

UC2 - Points transaction details

The database and mapper support transaction history through `PointsTransactionMapper.selectByUid`. Redemption and manual-award flows also create `PointsTransactionDO` records. `MyPointsVO` and `guest_my_points.jsp` show the complete signed transaction history in the Points History tab:

- transaction date;
- positive or negative points change;
- transaction type (`EARN` or `REDEEM`);
- related order ID, when applicable.

UC3 - Redeem products

The guest redemption flow is implemented by `GuestPointsServiceImpl`:

1. Load and validate the guest and product (including an in-stock pre-check).

2. Validate the chosen payment strategy.

3. Create a pending order.

4. Atomically deduct points with an optimistic balance check (compare-and-update).

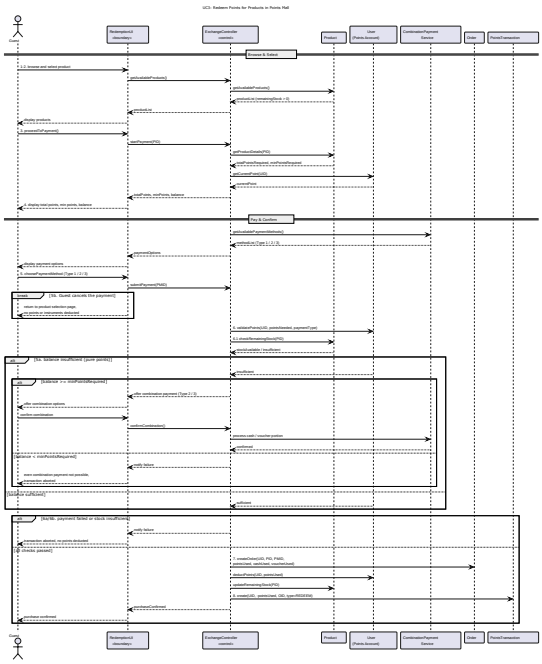
5. Settle the non-points portion through `PaymentGateway` (simulated cash charge or voucher redemption).

6. Mark the order as paid through the `OrderContext` state pattern with a compare-and-update status transition.

7. Insert a negative points transaction linked to the order.

8. Publish a `PointsDeductedEvent`; `RewardDistributionListener` runs synchronously within the same transaction and deducts one unit of product stock as the reward-distribution step.

The My Redemptions tab reads the same paid order records through `OrderMapper.selectByUid`, so points-only, cash, and voucher redemptions all appear after successful completion.



The supported payment strategies are:

- points only;
- points plus cash;
- points plus voucher.

The cash and voucher interactions are intentionally simulated for this academic prototype. Real external payment authorization, capture, refund, timeout reconciliation, and provider-failure handling are **out of scope**, not incomplete implementation work.

UC4 - Manage points rules

Managers can list, create, and update rules. The service validates rule type and the time range for period rules. Level rules clear the threshold because they do not use a minimum external order amount.

The rule data is stored and managed, but no external hotel-order earning process in this application reads the rules to calculate newly earned points. The management part of UC4 is therefore implemented; **the following calculation and award integration is not implemented**.

UC5 - Manually add points

Managers verify a target user, enter a positive point amount, and may optionally provide a reason. The request is sent to the persistent Artemis queue. `PointsAdjustmentConsumer` updates the balance and creates an `EARN` transaction. Exceptions cause message redelivery, which gives automatic retry behavior.

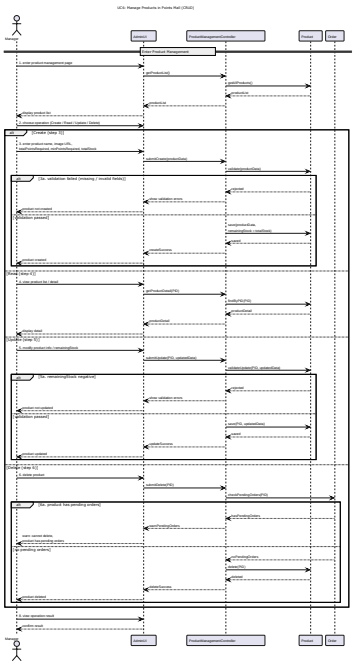
The optional reason is currently written to the application log, not persisted in `points_transaction`, whose schema has no reason column. **There is also no queue-status API, durable request ID, dead-letter administration, or user-facing retry status.**

UC6 - Manage products

Managers can:

- create a product;
- list products;
- get product details;
- update product information and stock;
- delete a product when it has no active orders.

On creation, `remainingStock` is set to `totalStock`. The service validates that minimum points do not exceed total points and that remaining stock does not exceed total stock. **The guest-facing catalogue currently filters unavailable products instead of showing a complete availability status.**



Required detailed use-case descriptions

The following two key use cases are the detailed use cases selected for the report. They follow the lecture-note template and are consistent with the implementation status above.

Key use case UC3 - Redeem points for a product

Field	Description
Primary actor	Guest
Supporting actors/systems	Spring Security, payment strategy, H2 database, synchronous reward listener
Goal	Exchange points, or points plus a simulated cash/voucher payment, for an available product
Trigger	An authenticated guest submits the redemption form or REST request
Preconditions	The guest is authenticated and owns the session; the product exists; the product is available; the request passes validation
Main success scenario	1. Load the guest, product, and payment method. 2. Validate the selected payment strategy. 3. Create a pending order. 4. Atomically deduct points. 5. Process the simulated non-points payment when applicable. 6. Transition the order to paid. 7. Insert the signed redemption transaction. 8. Publish `PointsDeductedEvent` and deduct one unit of stock.
Alternative and exception flows	Insufficient points, invalid payment details, missing product, unavailable stock, failed optimistic update, or payment failure abort the transaction and do not leave a partial redemption.
Postconditions	A successful redemption has a paid order, a negative points transaction linked to that order, and one fewer unit of remaining stock. A failed redemption leaves the protected balance, stock, and order data unchanged.
Quality attributes	Extensibility through `PaymentStrategy`; reliability through transactions and conditional updates; security through ownership and role checks.

Key use case UC6 - Manage products

Field	Description
Primary actor	Manager
Supporting actors/systems	Spring Security, `ProductManagementService`, `ProductMapper`, H2 database
Goal	Create, inspect, update, and delete points-mall products while preserving stock and order constraints
Trigger	An authenticated manager submits a product-management operation
Preconditions	The manager is authenticated with `ROLE_MANAGER`; create/update data passes DTO validation; the target product exists for read/update/delete operations
Main success scenario	1. Manager opens the product-management page. 2. The controller validates the request. 3. The service checks point and stock invariants. 4. The mapper persists the create/update/delete operation. 5. The updated product list or result is returned.
Alternative and exception flows	Invalid points or stock values are rejected; an unknown product returns an error; deletion is rejected when active orders reference the product.
Postconditions	A successful create or update stores a product with valid point and stock values; a successful delete removes only a product with no active orders.
Quality attributes	Security through manager-only routes; modifiability through controller/service/mapper separation; auditability through persisted product and order records.

4. Functional Architecture

The application follows a layered web architecture:

```
JSP pages / REST clients
          |
          v
Controllers
          |
          v
Application services
          |
          v
MyBatis mappers
          |
          v
H2 database
```

The asynchronous manual-award path is:

```
Manager UI
    |
    v
PointsAdjustmentController
    |
    v
PointsAdjustmentService
    |
    v
Artemis queue
    |
    v
PointsAdjustmentConsumer
    |
    +--> UserMapper
    |
    +--> PointsTransactionMapper
```

Architectural patterns

- **MVC + REST:** controllers receive HTTP requests and return JSON or forward to JSP

  views; DTOs, VOs, and DOs carry model data; services hold business logic; the web layer does not access the database directly.

- **Layered architecture (independently researched):** the request path is Controller

  → Service → MyBatis Mapper → H2. Domain objects are shared data objects, not an additional call-chain step. This pattern was selected from independent architecture research rather than from the lecture list of MVC, Broker, and Scheduler.

- **Event-driven architecture:** points-adjustment requests are placed on a persistent

  Artemis queue and processed later by a consumer. The producer and consumer share only the queue contract, allowing retry on processing conflicts and keeping the caller independent from the work.

Technology deployment pipeline

The system is intentionally **monolithic**: one Spring Boot application is built and packaged as one executable JAR. The runtime pipeline is:

```
Browser
  -> embedded Tomcat web/application server
  -> Spring MVC JSP controllers and REST controllers
  -> service layer
  -> MyBatis Data Mapper
  -> H2 relational EIS/database

Manager adjustment request
  -> Spring service
  -> embedded Artemis JMS message bus
  -> adjustment consumer
  -> service layer and H2 database
```

Maven and GitHub Actions form the delivery pipeline: a push runs compilation and tests, and a tag packages the executable JAR and creates a GitHub Release. The current academic project stops at packaging because no deployment server is configured. A production version would replace the embedded server/database/broker with managed equivalents while retaining the same application boundaries.

Main packages

Package	Responsibility
`controller`	REST endpoints and JSP page controllers
`service`	Application service interfaces
`service.impl`	Business rules and transaction orchestration
`domain`	Persistence/domain objects
`dto`	Validated request payloads
`vo`	View and API response objects
`mapper`	MyBatis data-access interfaces
`payment`	Payment strategy abstractions and implementations
`messaging`	Artemis producer, consumer, messages, and events
`order`	Order state abstraction
`security`	Authentication, login flow, and role handling
`config`	Spring Security and JMS configuration

---

5. Design Patterns

5.1 Layered architecture

Controllers do not directly implement persistence logic. For example:

```
ProductManagementController
        -> ProductManagementService
        -> ProductMapper
        -> ProductDO
```

This keeps HTTP concerns separate from business validation and database access.

5.2 Strategy pattern for payment

`PaymentStrategy` defines the validation contract. The concrete strategies are:

- `PointsOnlyStrategy`
- `PointsPlusCashStrategy`
- `PointsPlusVoucherStrategy`

`PaymentStrategyFactory` picks a strategy by `PaymentTypeEnum`. A new payment instrument can therefore be added as another strategy without changing the main redemption orchestration.

The class relationship and concrete implementations are shown above. This is not a Chain of Responsibility: there is no sequential handler chain, fallback handler, or `next` reference.

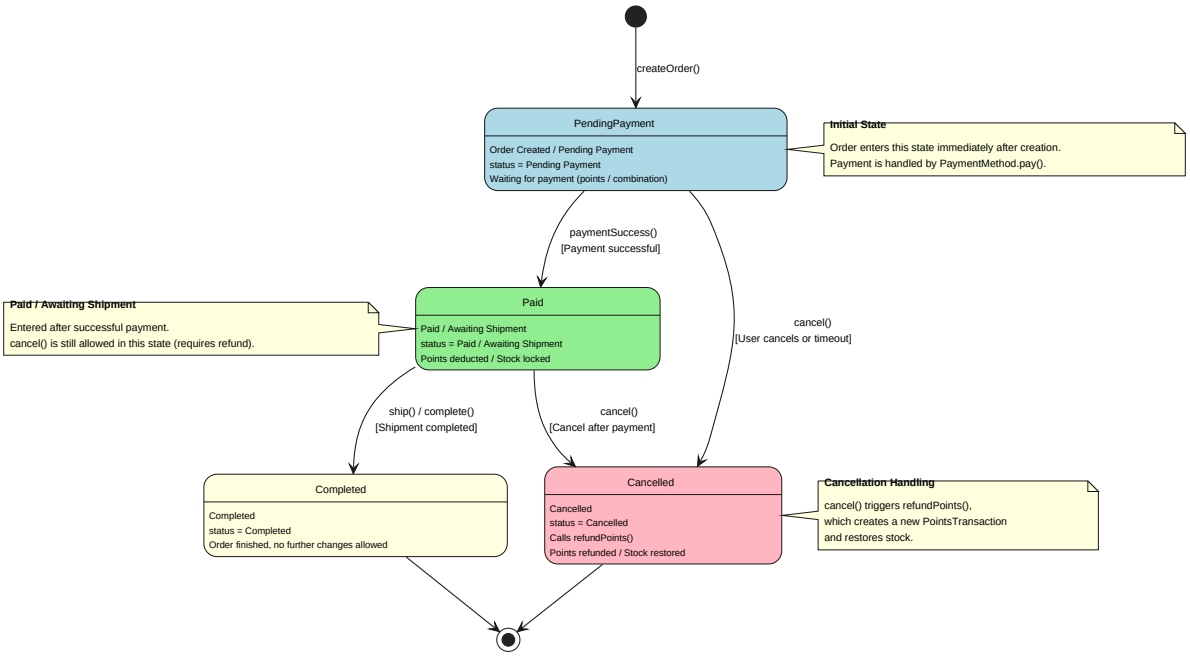
5.3 State pattern for orders

The `order` package contains:

- `OrderState`

- `PendingPaymentState`
- `PaidState`
- `CompletedState`
- `CancelledState`
- `OrderContext`

The current guest redemption service uses `OrderContext` to validate the pay transition, then persists the new status with a compare-and-update. The state classes provide the model for order lifecycle behavior and can be expanded as cancellation and fulfilment workflows are added.



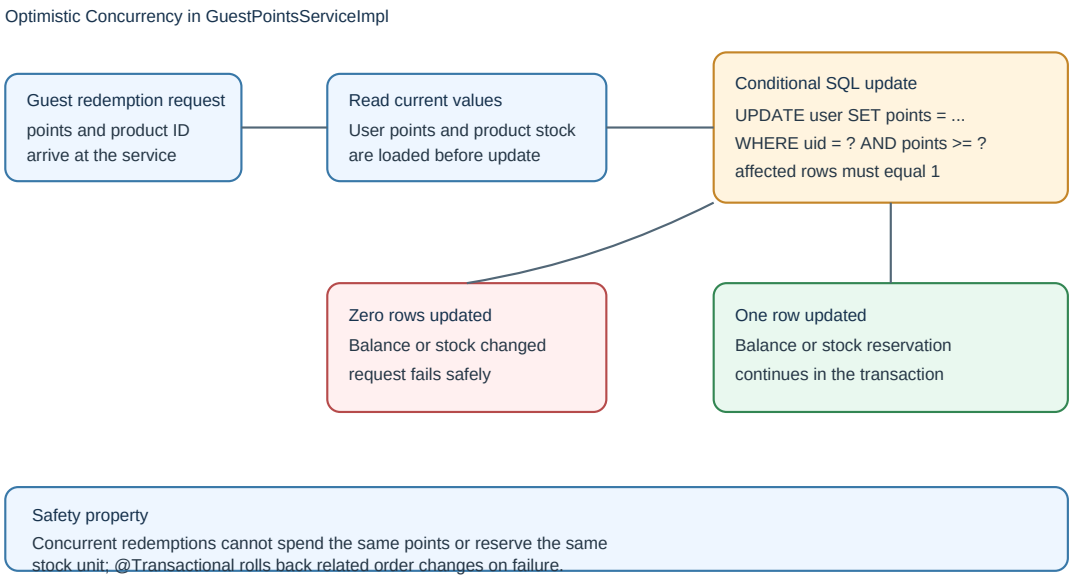
5.4 Message-driven processing

Manual point awards are decoupled from the manager request through Artemis JMS. The producer acknowledges the request by enqueueing it, while the consumer performs the balance update and transaction insert. An exception lets the broker redeliver the message.

The queue, producer, consumer, database writes, and retry path are shown above. Persistent delivery, listener rollback/redelivery, and optimistic locking provide the implementation's reliability mechanism; there is no claim of exactly-once processing beyond the current transaction and message configuration.

5.5 Optimistic concurrency

Point and stock updates use conditional database updates. A redemption fails if the stored balance or stock changed between the initial read and the update. This stops two concurrent requests from spending the same points or stock unit.

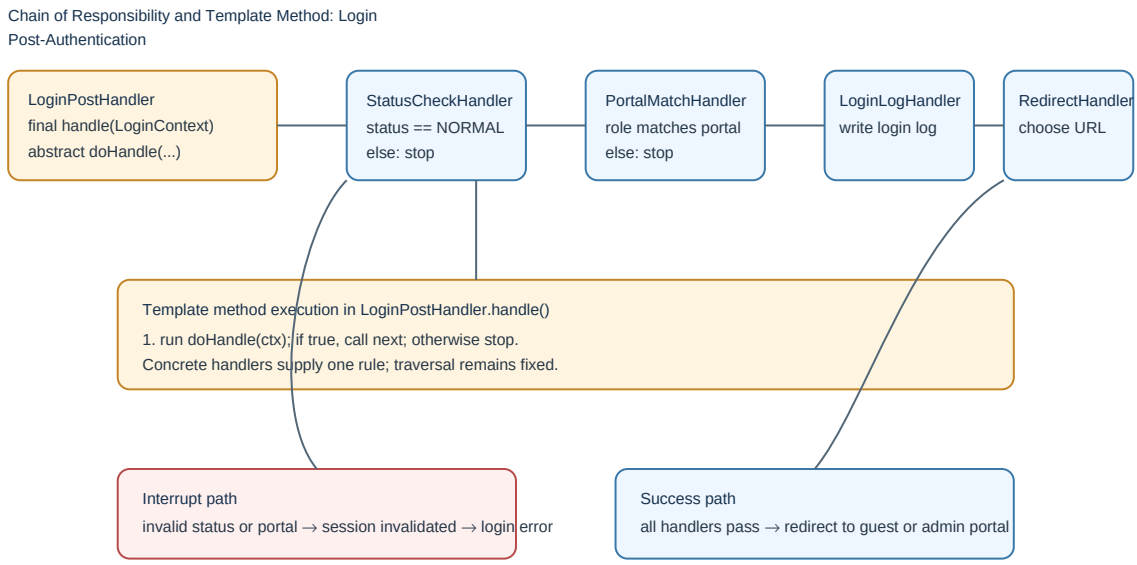


5.6 Chain of Responsibility for post-login checks

The security package contains a hand-written Chain of Responsibility for business checks that run after Spring Security has authenticated the password:

```
StatusCheckHandler
        -> PortalMatchHandler
        -> LoginLogHandler
        -> RedirectHandler
```

Each `LoginPostHandler` performs one check and passes the shared `LoginContext` to its successor. Returning `false` stops the chain. This is separate from Spring Security's framework `SecurityFilterChain`; the project-owned chain handles application-specific status, portal, logging, and redirect decisions.

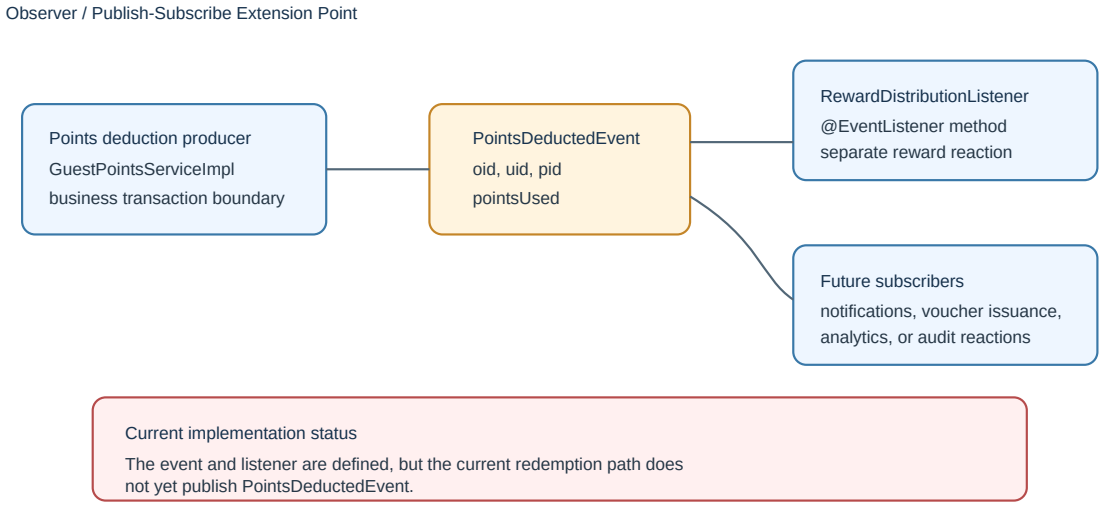


5.7 Template Method in login handlers

`LoginPostHandler.handle()` is a final template method. It runs the common control flow (run the current handler, then call the successor when the check passes), while concrete handlers implement the variable `doHandle()` step. This keeps chain traversal consistent and limits each handler to one business rule.

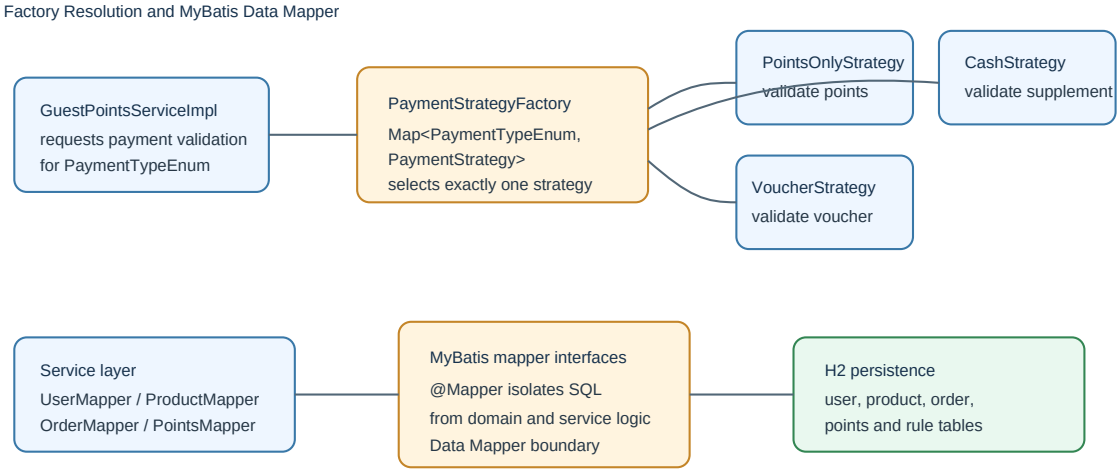
5.8 Observer / publish-subscribe event handling

`RewardDistributionListener` and `PointsDeductedEvent` provide an application-level Observer / publish-subscribe extension point. The redemption flow deducts points, settles payment, and creates the order; the listener then deducts one unit of product stock as the reward-distribution step. Spring's `@EventListener` runs synchronously inside the publisher's transaction, so the stock deduction participates in the same redemption transaction and atomicity is preserved while reward logic stays decoupled from the point-deduction service. Additional listeners, such as notification or voucher issuance, can subscribe to the same event without changing the existing redemption code.



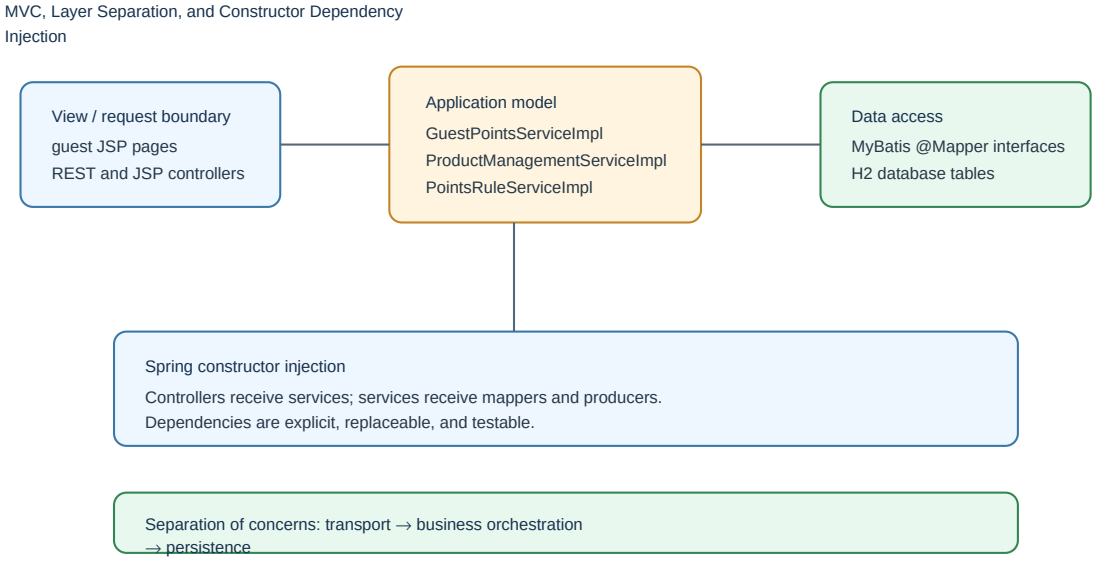
5.9 Factory and Data Mapper patterns

`PaymentStrategyFactory` is a Factory object that resolves a payment strategy from `PaymentTypeEnum`. The MyBatis mapper interfaces (`UserMapper`, `ProductMapper`, `OrderMapper`, and others) implement the Data Mapper pattern: persistence operations are kept outside the domain objects and services.



5.10 MVC and dependency injection

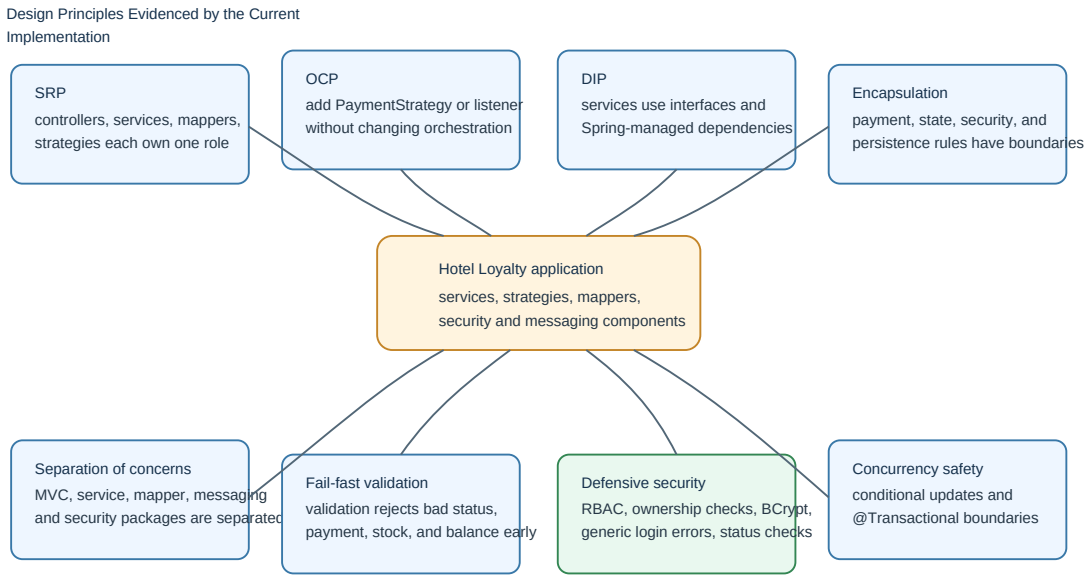
The JSP controllers and REST controllers provide the Model-View-Controller separation: controllers handle requests, services provide application behavior, and JSP pages or `Result` objects provide the view/response. Constructor injection is used throughout the controllers, services, producers, consumers, and listeners, making dependencies explicit and testable.



5.11 Repository-level design principles

The implementation applies these principles:

- **Single Responsibility Principle:** controllers handle transport, services orchestrate business rules, mappers handle persistence, and payment strategies validate one payment type.
- **Open/Closed Principle:** new payment strategies and event listeners can be added without rewriting the redemption orchestration or existing listeners.
- **Dependency Inversion Principle:** services depend on mapper and producer interfaces or abstractions rather than constructing database or messaging infrastructure directly.
- **Encapsulation:** payment validation, order transitions, security checks, and persistence operations are kept behind focused interfaces and service boundaries.
- **Separation of concerns:** authentication, authorization, business validation, persistence, messaging, and presentation are placed in separate packages and layers.
- **Fail-fast validation:** invalid status, portal, payment, stock, balance, and input conditions are rejected before dependent operations continue.
- **Defensive security design:** role-based URL rules, ownership checks, BCrypt password hashing, generic login errors, and explicit account-status checks reduce unauthorized access and information disclosure.



6. Domain Model

The core persistent entities are:

Entity	Purpose
`UserDO`	Guest or manager identity, role, status, level, and point balance
`ProductDO`	Points-mall product and stock information
`PaymentMethodDO`	Configured payment method catalogue
`OrderDO`	A guest redemption request and its payment data
`PointsTransactionDO`	Signed points audit record
`PointsRuleDO`	Manager-maintained earning rule definition

The current implementation stores both guest and manager accounts in `UserDO`. They are not Java subclasses. Spring Security maps the stored role to `ROLE_GUEST` or `ROLE_MANAGER`.

The main relationships are:

```
User 1 ---- * Order
User 1 ---- * PointsTransaction
Product 1 ---- * Order
PaymentMethod 1 ---- * Order
Order 1 ---- * PointsTransaction
PointsRule ..> User
```

Conceptual/initial design sketches

The analysis-stage design evidence is kept separate from the implementation description:

- **Candidate objects:** `candidate_objects_1.0.xlsx`

identifies candidate actors, entities, boundary objects, control objects, and supporting objects before implementation.

- **Class sketch:** `class_diagram_1.0.svg` shows the principal

classes, interfaces, visibility, inheritance, aggregation/composition, associations, and dependencies. The final implementation uses role-based `UserDO` rather than separate `Guest` and `Manager` subclasses; this is documented as an implementation difference.

- **Interaction sketches:** `redeem_points_for_products.svg`

and `manage_products.svg` are the sequence diagrams for the two detailed key use cases.

- **State sketch:** `order_state_diagram_1.0.svg` gives

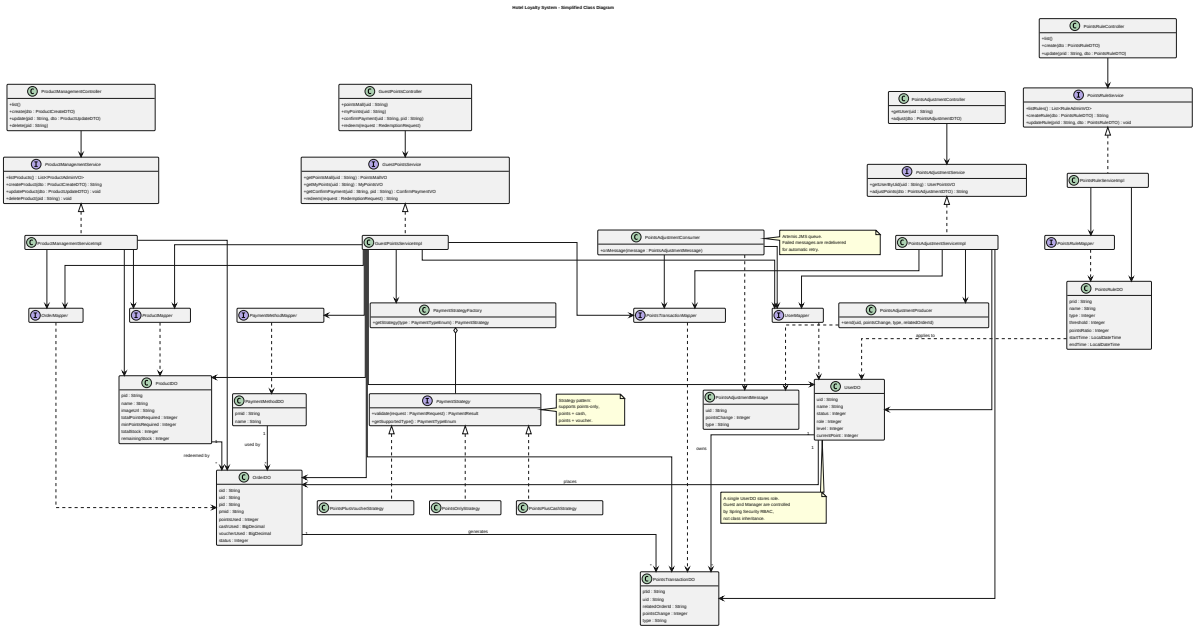
annotated order transitions and transition strings.

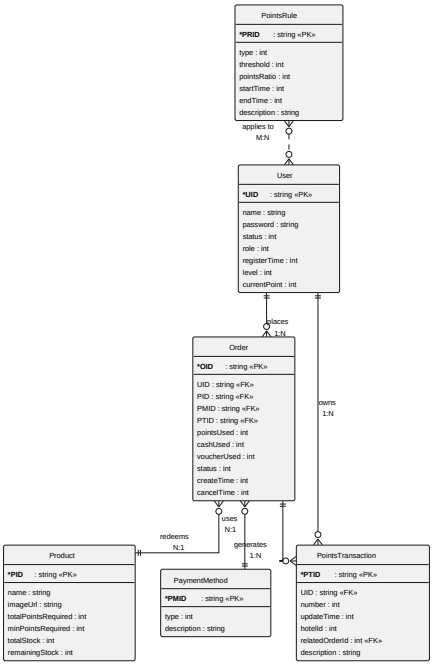
- **Entity relationship sketch:** `entry_relationship_diagram.svg`

shows the relational entities and cardinalities.

The key interface contracts used by the sketches are:

Interface/operation	Precondition	Postcondition
`PaymentStrategy.validate(request)`	Payment type and request values are present	Returns a valid payment result or rejects the request without changing persistent state
`ProductManagementService.create/update`	Manager authorization and valid point/stock invariants	A product is stored with `0 <= remainingStock <= totalStock`
`GuestPointsService.redeem`	Authenticated owner, available product, and valid payment request	Success creates a paid order, signed transaction, and stock deduction atomically; failure preserves protected state
`OrderState.pay(context)`	Order is in the pending-payment state and payment validation succeeded	Order transitions to paid, or the operation is rejected without a partial transition





The application uses business identifiers (`uid`, `pid`, `pmid`, `oid`, `ptid`, and `prid`) in service and mapper operations. The database schema intentionally keeps foreign-key enforcement in the application layer.

7. Database Design

The database is H2 in file mode:

```
./data/hotelloyalty
```

Schema initialization is enabled on startup. The main tables are:

```
user
product
payment_method
order
points_transaction
points_rule
```

Important data rules:

- all tables have a generated numeric `id`;
- business identifiers have unique constraints;
- `points_change` is signed;
- `remaining_stock` is maintained separately from `total_stock`;
- `related_order_id` is nullable for non-order earnings;
- orders and points transactions keep creation and modification timestamps.

The schema is defined in `schema.sql`, and initial demonstration data is in `data.sql`.

---

8. API and Page Endpoints

Authentication

Method	Endpoint	Purpose
GET	`/login`	Guest login page
GET	`/admin/login`	Manager login page
POST	`/auth/login`	Spring Security login processing
POST	`/logout`	Session logout

Guest pages and APIs

Method	Endpoint	Purpose
GET	`/guest/{uid}/account`	Render the guest account page
GET	`/guest/{uid}/points-mall`	Render the guest points mall
GET	`/guest/{uid}/my-points`	Render balance and redemption page
GET	`/guest/{uid}/confirm-payment/{pid}`	Render payment confirmation
POST	`/guest/{uid}/redemptions`	Submit JSP redemption
GET	`/api/guest/{uid}/account`	Get account data
GET	`/api/guest/{uid}/points-mall`	Get mall data
GET	`/api/guest/{uid}/my-points`	Get guest points data
GET	`/api/guest/{uid}/confirm-payment/{pid}`	Get payment data
POST	`/api/guest/redemptions`	Submit REST redemption

Guest endpoints verify that the URL or request UID matches the logged-in session identity.

Manager pages and APIs

Method	Endpoint	Purpose
GET	`/admin/products`	Product management page
GET	`/admin/rules`	Points-rule management page
GET	`/admin/points`	Manual points adjustment page
GET	`/api/admin/products`	List products
POST	`/api/admin/products`	Create product
GET	`/api/admin/products/{pid}`	Get product
PUT	`/api/admin/products/{pid}`	Update product
DELETE	`/api/admin/products/{pid}`	Delete product
GET	`/api/admin/rules`	List rules
POST	`/api/admin/rules`	Create rule
GET	`/api/admin/rules/{prid}`	Get rule
PUT	`/api/admin/rules/{prid}`	Update rule
GET	`/api/admin/points/user/{uid}`	Verify target user
POST	`/api/admin/points/adjustment`	Enqueue manual point adjustment

All `/admin/**` and `/api/admin/**` routes require the manager role.

---

9. Security and Reliability

Authentication and authorization

- Session-based Spring Security authentication.
- BCrypt password encoding.
- Separate guest and manager login portals.
- URL-level role-based access control.
- Guest ownership checks against `SecurityContext`.
- Explicit 30-minute session timeout.

Data and transaction safety

- Redemption is transactional.
- Point balance updates use compare-and-update semantics.
- Stock deduction checks remaining stock in the update operation.
- Product deletion checks for active orders before deletion.
- Validation annotations are used on request DTOs.

Asynchronous reliability

- Embedded Artemis is configured as persistent.
- Manual adjustments are queued before processing.
- Consumer exceptions trigger broker redelivery.
- Point adjustment and transaction creation are performed by the consumer.

Current limitations

- CSRF is disabled because the current JSP JavaScript calls REST endpoints without CSRF tokens. This is acceptable for the academic prototype but should be fixed for production.
- No user-facing queue status or dead-letter workflow is exposed.
- The adjustment reason is logged but not stored in the database.
- The points-rule table is not yet connected to an external hotel-order earning flow.

---

10. User Interface

The UI is server-rendered using JSP and JSTL.

The two required design-time GUI prototypes are:

- `guest_prototype_1.1.html`, covering guest login,

account, points mall, and redemption flow.

- `admin_prototype_1.1.html`, covering manager login,

product management, points rules, and manual point adjustment.

The supporting prototype pages `points_mall_prototype_1.1.html`, `my_points_prototype_1.1.html`, and `confirm_payment_prototype_1.1.html` provide the detailed guest screens.

View	Purpose
`login.jsp`	Guest login
`admin_login.jsp`	Manager login
`guest_account.jsp`	Show account information
`guest_points_mall.jsp`	Browse products
`guest_confirm_payment.jsp`	Select and confirm payment
`guest_my_points.jsp`	Show balance and redemption records
`admin_products.jsp`	Product CRUD
`admin_points_rules.jsp`	Points rule CRUD
`admin_points_adjust.jsp`	Manual point award
`products.jsp`	Product-list demonstration page

The working demonstration page is:

```
http://localhost:8080/demo/products
```

The manager product page is:

```
http://localhost:8080/admin/products
```

---

11. Demo Data and Running the Application

Prerequisites

- JDK 21 or a compatible newer JDK
- Maven Wrapper included in the project

Start the application

From the application directory:

```
cd Project/hotel-loyalty
./mvnw spring-boot:run
```

On Windows:

```
cd Project\hotel-loyalty
mvnw.cmd spring-boot:run
```

The default server address is:

```
http://localhost:8080
```

Build and test

```
cd Project/hotel-loyalty
./mvnw test
./mvnw package
```

The generated executable JAR is placed under `Project/hotel-loyalty/target/`.

Demonstration credentials

The seed data defines:

Role	Username	Password
Manager	`admin`	`admin123`

Role	Username	Password
Guest	`guest`	`guest123`

These credentials are for local demonstration only and must be replaced in a deployed environment.

Useful local tools

- H2 Console: `http://localhost:8080/h2-console`
- OpenAPI JSON: `http://localhost:8080/v3/api-docs`
- Swagger UI: `http://localhost:8080/swagger-ui.html`

For the H2 console, use the JDBC URL configured in `application.yaml` and the local `sa` user.

---

12. Testing Status

The repository currently contains a Spring context smoke test:

```
src/test/java/com/hotel/hotel_loyalty/HotelLoyaltyApplicationTests.java
```

The current automated suite has 103 tests covering:

1. Login, account-status, guest-ownership, and manager-role security behavior.
2. Product create, update, delete, validation, and active-order protection.
3. Points-only, points-plus-cash, and points-plus-voucher validation.
4. Minimum-point, supplement-conversion, insufficient-balance, and out-of-stock redemption.
5. Atomic balance, stock, order, and points-transaction changes.
6. Voucher redemption transaction-history linkage.
7. Manual-award queue processing and optimistic-lock retry behavior.
8. Event-driven reward distribution: `PointsDeductedEvent` publication and listener stock-deduction behavior.
9. Mapper, controller, service, state, and application-context behavior.

The suite does not yet provide browser-level end-to-end coverage for every use case, real external payment-provider tests, points-rule calculation integration, or queue dead-letter operations.

The full suite currently passes with `./mvnw clean test`.

Testing approach

The project uses JUnit 5 and a testing-pyramid approach organized by distance from the business core. The intended approximate distribution is 70% unit tests, 20% integration tests, and 10% end-to-end or system-level tests. Unit tests are fast and cover most branches; the slower integration tests are reserved for critical paths such as the message flow, persistence, and Spring wiring. Browser-level end-to-end coverage remains an identified gap.

CI/CD

CI/CD has been implemented with GitHub Actions. Pushes run the Maven verification/test pipeline, and tagged releases package the application JAR and create a GitHub Release. Because the project has no deployment server, continuous delivery currently ends at automated packaging; no server deployment is performed.

---

13. UML and Supporting Documents

The repository contains these supporting artifacts:

Artifact	Location
Use-case catalogue	`Use_case_1.0.xlsx`
Detailed UC3 and UC6 descriptions	`Use_Case_Description_UC3_UC6_v1.1.xlsx`
Original class diagram	`class_diagram_1.0.svg`
Use-case diagram	`all_user_case.svg`
Product-management sequence diagram	`manage_products.svg`
Redemption sequence diagram	`redeem_points_for_products.svg`
Order state diagram	`order_state_diagram_1.0.svg`

Artifact	Location
Entry relationship diagram	`entry_relationship_diagram.svg`
Layered package architecture	`layered_architecture.svg`
Payment strategy UML	`payment_strategy_uml.svg`
Message-driven points adjustment	`message_driven_points_adjustment.svg`
Optimistic concurrency control	`optimistic_concurrency.svg`
Login Chain of Responsibility and Template Method	`login_chain_template.svg`
Observer event extension point	`observer_event.svg`
Factory and Data Mapper	`factory_data_mapper.svg`
MVC and dependency injection	`mvc_dependency_injection.svg`
Design principles evidence	`design_principles.svg`
Database schema	`schema.sql`

The latest assignment documents used to update this report are:

Topic	Source document
Software lifecycle	`3.2 lifecycle discussion`
System architecture	`5.0 architecture discussion`
Package/class/LOC evidence	`7.1`, `7.2`, `7.3`
MVC and architectural patterns	`8.1`
Automated testing	`8.4`
CI/CD	`9.1`
Four design patterns	`Four Design Patterns Implemented`
Generative-AI prompt list	`Prompt list`
Design and implementation critique	`Critique`

The original UML assets describe the conceptual design and should be read together with the current implementation notes in this report. In particular:

- the running code uses one `UserDO` with role-based security rather than Java `Guest` and `Manager` subclasses;
- payment methods are stored as database records while payment validation is implemented through `PaymentStrategy`;
- the current guest page shows both redemption records and the full points transaction history;
- order-state classes exist; the redemption service uses them to validate the pay transition, and cancellation/fulfilment workflows remain to be connected.

---

14. Implementation Gaps and Recommended Next Steps

Reviewed against the current code on 27 October 2026. All four gaps below are still open.

Priority 1: **[Key gap]** Connect points rules to earning

`PointsRuleServiceImpl` only performs CRUD on rules, and `PointsRuleMapper.selectByTypeAndThreshold` is not called by any main-code flow. Introduce an external-order earning service that picks an applicable `PointsRuleDO`, calculates the points from the ratio, and sends the resulting earning event through the existing adjustment/message mechanism.

Priority 2: **[Key gap]** Add operational queue management

`PointsAdjustmentMessage` carries no reason and no durable request ID, `points_transaction` has no reason column, the reason is only written to the application log, and `JmsConfig` defines no dead-letter queue. Persist the manual-adjustment reason, add a durable request identifier, and expose queued, processed, retrying, and permanently failed states with dead-letter administration.

Priority 3: **[Partially complete]** Make product availability explicit

`ProductVO` only exposes `pid`, `name`, `imageUrl`, and `totalPointsRequired`; `getPointsMall` filters products through `selectAvailable()`. The backend filters out-of-stock products, but the guest-facing model should expose availability and stock status explicitly where the use case requires it.

Priority 4: **[Partially complete]** Complete order lifecycle parity

The redemption service now validates the pay transition through `OrderContext` and persists it with a compare-and-update, but `OrderContext.cancel()` and `complete()` are not called by any workflow. Connect the existing order-state classes to cancellation, fulfilment, and shipment workflows. UC1 account viewing is complete; the remaining gap is order lifecycle behavior, not account data.

Test coverage status

The current automated suite contains 103 tests, all passing with `./mvnw clean test`; the covered areas are listed in section 12.

---

15. Conclusion

The current project is a working academic prototype of a hotel loyalty points mall. Its strongest implemented areas are the core redemption transaction, role-based manager/guest separation, product CRUD, strategy-based payment validation, and asynchronous manual point adjustment.

The main differences between the original design documents and the current code are implementation-level rather than architectural: the application uses role-based persistence instead of user inheritance, strategy classes instead of payment-method subclasses, and service/mapper layers instead of domain objects containing all behavior.

The next meaningful step is to show product availability more explicitly and connect managed points rules to an actual points-earning workflow. Once those are implemented and covered by browser-level tests, the project will have much stronger traceability from the use-case catalogue through the UML artifacts to the running application.

---

16. Technology Stack and Technical References

This section identifies the technologies used by the implemented system and the role of each technology in the solution.

Area	Technology	Use in this project
Language/runtime	Java 21	Application source, domain objects, services, tests, and build target
Application framework	Spring Boot 4.0.8	Application bootstrapping, auto-configuration, dependency management, and embedded server
Web layer	Spring MVC / WebMVC	REST controllers, page controllers, request binding, validation, and exception handling
Presentation	JSP, JSTL, HTML, CSS, JavaScript	Server-rendered guest and manager pages and browser-side API calls
Security	Spring Security, BCrypt	Session authentication, guest/manager role-based authorization, password hashing, and ownership checks
Persistence	MyBatis 4.0.1	Data Mapper interfaces and SQL-backed access to the application tables
Database	H2 Database	Local file-mode relational database for development, demonstration, and automated tests
Messaging	Jakarta Messaging (JMS), embedded Apache ActiveMQ Artemis	Persistent queue and asynchronous processing of manager point adjustments
API documentation	Springdoc OpenAPI 3.1.1, Swagger UI	OpenAPI 3 document and interactive API documentation
Validation	Jakarta Bean Validation / Spring Boot validation	DTO constraints and fail-fast request validation
Build	Apache Maven and Maven Wrapper	Reproducible dependency resolution, compilation, testing, and packaging
Testing	JUnit 5, Spring Boot test support, Spring Security Test, MyBatis test support	Unit, integration, controller, security, service, mapper, and context tests
Code generation	Lombok	Reduced boilerplate for selected Java model and service classes
Modelling	UML/SVG, Excel use-case and candidate-object workbooks	Requirements, class, sequence, state, component, deployment, package, and data-model communication
Version control	Git and GitHub	Source history, collaboration, traceability, and version-control evidence

The implementation is intentionally a single deployable Spring Boot application. H2 and embedded Artemis keep the academic prototype self-contained; they are replaceable infrastructure choices rather than business-domain dependencies.

Technology and theory references

The implementation and design discussion are grounded in the five references listed in Section 25. The correspondence is:

- **Architecture and quality attributes:** Bass, Clements, and Kazman.

- **Strategy, State, Observer, Factory, and Chain of Responsibility:** Gamma et al.
- **MVC, Data Mapper, and transaction-script/service-layer separation:** Fowler.
- **Spring MVC, dependency injection, Spring Security, and Spring Boot:** Walls.
- **Agile lifecycle, principles, and the complexity/agility trade-off:** Martin.

The online framework documentation used for implementation verification is linked in the technology table where appropriate; it is not counted as part of the formal five-item reference list.

---

17. Assignment Deliverables and Traceability Matrix

The following matrix converts the supplied assignment workbook into a report-ready submission checklist. “Evidence” links to the current repository artifact; “status” is the state of the evidence in this repository, not a claim that every optional extension has been implemented.

Assignment item	Required report output	Evidence in this repository	Status
3.1 Business scenario	Maximum one-page narrative	Section 1; `3.1 business scenario`	Complete
3.2 Software lifecycle	Maximum two-page lifecycle discussion and complexity/agility tension	`3.2 lifecycle discussion`; Section 18	Complete
3.3 Project plan	Scope, ownership, schedule, dependencies, and completion evidence	`Project plan`, `final plan screenshot`; Section 19	Complete
4.1 Requirements	Use-case diagram and two detailed key use cases	`Use-case diagram`; `UC3 and UC6 descriptions`; Section 3	Complete
4.2 Quality attributes	At least one “-ility” must be extensibility, plus quality tactics	Section 20; Sections 5 and 9	Complete
4.3 GUI prototypes	Two GUI prototypes and explanation of user flow	`GUI prototypes`; actual running screenshots in Section 10	Complete
5.0 Architecture	Maximum two-page architecture discussion, two architectural patterns, and a high-level package view	`Architecture discussion`; Sections 4 and 5; `package diagram`	Complete
6.1–6.5 Design sketches	Candidate objects, class, sequence/communication, state, and ER diagrams	`Candidate objects`; `UML assets`; Section 13	Complete
7.1–7.4 Transparency	Package/class counts, authorship and LOC, team contribution, patterns/testing/CI-CD ownership	`7.1 package counts`, `7.2 class authorship and LOC`, `7.3 team LOC`; Section 21	Complete for the archived commit
8.1–8.4 Code evidence	MVC/REST, four patterns, key use cases, and automated testing snippets	`8.1 MVC and architectural patterns`, `four design patterns`, `8.4 automated testing`, Sections 5, 8, and 12; `src/main/java`; `src/test/java`	Complete
8.5 Version control	Visual evidence of version-control use	Git repository: <https://github.com/fordringd2008/CS6451-Advanced-software-design>	Complete; add screenshot if required by marking rubric
9 Added value	Additional design or implementation value	Sections 5.4–5.10, 9, and 20; `9.1 CI/CD`	Complete
10 Recovered architecture	Design-time package, component, and deployment blueprints	`package diagram`, `component diagram`, `deployment diagram`	Complete
11 Critique	Maximum one-page critique of design and implementation quality	`Critique`; Section 22	Added
12 Reflection	A section entitled “Reflection” with a learning account for each team member	Section 23	Fordring drafted; Franklin entry requires confirmation
13 AI list	List of prompts and generative-AI platforms used	`Prompt list`; Section 24	Complete for the archived prompt list
14 References	Academic and technical references	Section 16 and Section 25	Added

For final export, the report should retain sections 1–25 in order and include the linked SVG diagrams as figures or an appendix. The assignment page limits apply to the named subsections, not to the evidence appendix.

---

18. Software Lifecycle Discussion

The project adopted a **Waterfall lifecycle**. Requirements were confirmed and frozen at the start, and the chance of later changes was considered low. Feedback was collected at fixed review points: requirements were reviewed in Week 5, design was completed during Weeks 5–6, and coding began from Week 7. A two-person team could communicate directly, so formal Agile ceremonies such as daily stand-ups would have added cost without enough benefit.

The team considered RUP but did not adopt it. RUP repeats analysis and design through multiple iterations and is better suited to large or high-risk systems requiring formal tracking and audits. This project has stable requirements, two team members, and no external audit requirement, so RUP's additional process would add effort without proportionate value.

The main tension was between **management of complexity and agility**. For this small, simple project, heavy documentation and strict process would slow delivery while adding little practical value. The team resolved the tension with a lightweight Waterfall approach: work in a clear sequence, keep only the communication and records needed for the assignment, and avoid process overhead that does not address a real project risk.

---

19. Project Plan Summary

The project plan divides work across requirements, architecture, GUI, coding, testing, documentation, and final review. The dependency order used by the project is:

```
Business scenario and use cases
        -> candidate objects and GUI prototypes
        -> architecture and UML
        -> implementation
        -> automated tests and traceability
        -> critique, reflection, AI list, and final references
```

The implementation evidence is organized around six use cases: UC1 login/account, UC2 points history, UC3 product redemption, UC4 points rules, UC5 manual point award, and UC6 product management. UC3 and UC6 are the two key use cases with detailed spreadsheet descriptions. The current schedule and named ownership remain in `-v1.0-3-10-2026.xlsx`; this README supplies the narrative that connects the plan to the delivered artifacts.

Roles, responsibilities, deadlines, and deliverables

The final plan assigns work by deliverable and uses Week 5–Week 8 milestones:

Team member	Role and contribution	Main deadlines	Deliverables
Fordring	Requirements lead, use-case analyst, UML/sketch contributor, implementation contributor, and final-report contributor	Week 5: scenario, use cases, quality attributes; Week 6: class/state/ER sketches; Week 7: coding and version-control evidence; Week 8: references and final review	Business scenario, use-case catalogue and key-use-case evidence, quality-attribute evidence, candidate objects, class/state/ER sketches, coding contribution, final references
Franklin	Lifecycle and architecture lead, GUI contributor, transparency/testing/CI-CD contributor, implementation contributor, and critique contributor	Week 5: lifecycle, GUI, architecture; Week 7: package/class/LOC evidence, patterns, automated testing, CI/CD, critique, coding; Week 8: reflection	Lifecycle discussion, GUI prototypes, architecture and package evidence, transparency tables, pattern/testing/CI-CD evidence, critique, coding contribution, reflection
Both	Joint implementation, integration, review, and final consistency checking	Weeks 7–8	Working Spring Boot application, tests, final report, and submission review

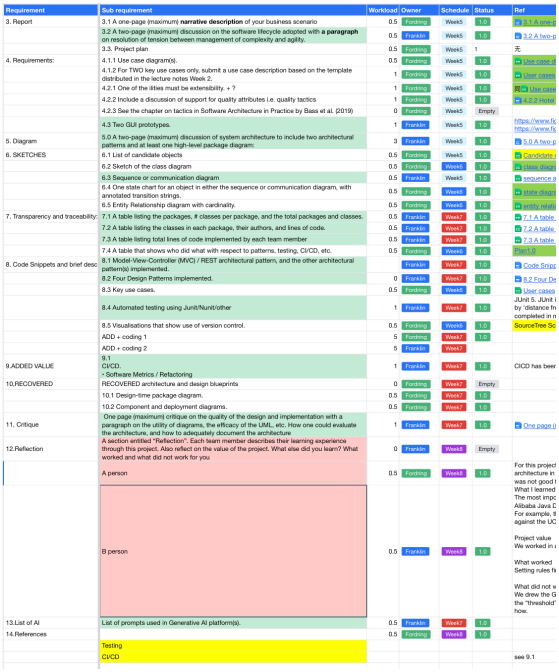
The plan is iterative at the evidence level: requirements and sketches were reviewed before implementation, implementation findings were fed back into the UML and gap notes, and tests and the final critique were used to revise the report before export.

Industry experience

The supplied project files do not contain verified employment or industry-experience details for either team member. The required table is included below so that the official information can be added without inventing credentials:

Team member	Industry domain(s)	Programming languages	Frameworks/tools	Evidence/status
Fordring	Not documented in the repository	Java, HTML/CSS/JavaScript are evidenced by this project	Spring Boot, Spring MVC, MyBatis, JUnit, Git/GitHub	Project experience evidenced; external industry experience requires confirmation
Franklin	Not documented in the repository	Java, HTML/CSS/JavaScript are evidenced by this project	Spring Boot, Spring MVC, MyBatis, JUnit, GitHub Actions	Project experience evidenced; external industry experience requires confirmation

Final project plan



Requirement	Sub requirement	Workload	Owner	Schedule	Status	Ref
3. Report	3.1 A one-page (maximum) **narrative description** of your business scenario	0.5	Fordong	Week5	1.0	3.1 A one-p
	3.2 A two-page (maximum) discussion on the software lifecycle adopted with **a paragraph** on resolution of tension between management of complexity and agility.	0.5	Franklin	Week5	1.0	3.2 A two-p
	3.3. Project plan	0.5	Fordong	Week5	1	无
4. Requirements:	4.1.1 Use case diagram(s).	0.5	Fordong	Week5	1.0	User cases
	4.1.2 For TWO key use cases only, submit a use case description based on the template distributed in the lecture notes Week 2.	1	Fordong	Week5	1.0	User cases
	4.2.1 One of the -ilities must be extensibility. + ?	1	Fordong	Week5	1.0	无 User case
	4.2.2 Include a discussion of support for quality attributes i.e. quality tactics	1	Fordong	Week5	1.0	4.2.2 Hotel
	4.2.3 See the chapter on tactics in Software Architecture in Practice by Bass et al. (2019)	0	Fordong	Week5	Empty	
	4.3 Two GUI prototypes.	1	Franklin	Week5	1.0	https://www.fig https://www.fig
5. Diagram	5.0 A two-page (maximum) discussion of system architecture to include two architectural patterns and at least one high-level package diagram:	3	Franklin	Week5	1.0	5.0 A two-p
6. SKETCHES	6.1 List of candidate objects	0.5	Fordong	Week5	1.0	Candidate
	6.2 Sketch of the class diagram	0.5	Fordong	Week6	1.0	class diagr
	6.3 Sequence or communication diagram	0.5	Franklin	Week5	1.0	sequence a
	6.4 One state chart for an object in either the sequence or communication diagram, with annotated transition strings.	0.5	Fordong	Week6	1.0	state diagra
	6.5 Entity Relationship diagram with cardinality.	0.5	Fordong	Week6	1.0	entity relati
7. Transparency and traceability:	7.1 A table listing the packages, # classes per package, and the total packages and classes.	0.5	Franklin	Week7	1.0	7.1 A table
	7.2 A table listing the classes in each package, their authors, and lines of code.	0.5	Franklin	Week7	1.0	7.2 A table
	7.3 A table listing total lines of code implemented by each team member	0.5	Franklin	Week7	1.0	7.3 A table
	7.4 A table that shows who did what with respect to patterns, testing, CI/CD, etc.	0.5	Fordong	Week8	1.0	Part7.R
8. Code Snippets and brief desc	8.1 Model-View-Controller (MVC) / REST architectural pattern, and the other architectural pattern(s) implemented.		Franklin	Week7	1.0	Code Snipp
	8.2 Four Design Patterns implemented.	0	Franklin	Week7	1.0	8.2 Four De
	8.3 Key use cases.	0.5	Fordong	Week6	1.0	User cases
	8.4 Automated testing using Junit/Nunit/other	1	Franklin	Week7	1.0	JUnit 5. JUnit by `distance` fi completed in n
	8.5 Visualisations that show use of version control.	0.5	Fordong	Week8	1.0	SourceTree Sc
	ADD + coding 1	5	Fordong	Week7	1.0	
	ADD + coding 2	5	Franklin	Week7		
9.ADDED VALUE	9.1 CI/CD. - Software Metrics / Refactoring	1	Franklin	Week7	1.0	CICD has beer
10.RECOVERED	RECOVERED architecture and design blueprints	0	Fordong	Week7	Empty	
	10.1 Design-time package diagram.	0.5	Fordong	Week7	1.0	
	10.2 Component and deployment diagrams.	0.5	Fordong	Week7	1.0	
11. Critique	One page (maximum) critique on the quality of the design and implementation with a paragraph on the utility of diagrams, the efficacy of the UML, etc. How one could evaluate the architecture, and how to adequately document the architecture	1	Franklin	Week7	1.0	One page c
12.Reflection	A section entitled "Reflection". Each team member describes their learning experience through this project. Also reflect on the value of the project. What else did you learn? What worked and what did not work for you	0	Franklin	Week8	Empty	
	A person	0.5	Fordong	Week8	1.0	For this project architecture in was not good I What I learned The most impo Alibaba Java D For example, t against the UC
	B person	0.5	Franklin	Week8	1.0	Project value We worked in a What worked Setting rules fi What did not w We drew the G the "threshold" how.
13.List of AI	List of prompts used in Generative AI platform(s).	0.5	Franklin	Week7	1.0	
14.References		0.5	Fordong	Week8	1.0	
	Testing					
	CI/CD					see 9.1

This image is the final project-plan snapshot and complements the underlying workbook.

---

20. Quality Attributes and Quality Tactics

The project treats the following qualities as architectural concerns:

Quality attribute	Tactics and implementation evidence
Extensibility	Strategy interfaces for payment, state objects for order lifecycle, event listeners for reward distribution, mapper/service interfaces, and dependency injection
Security	Session authentication, role-based URL rules, BCrypt, generic login errors, explicit account-status checks, and guest ownership validation
Reliability	Transactional redemption, conditional balance/stock updates, persistent Artemis messaging, retry on message-processing conflicts, and failure-visible implementation-gap documentation
Modifiability	Layered separation of controllers, services, mappers, and views; focused DTOs/VOs; constructor injection; isolated payment and messaging abstractions
Performance	Database-side conditional updates, filtered available-product queries, and asynchronous manual point adjustment
Testability	Constructor injection, service boundaries, mapper abstractions, Spring test slices, security test support, and a test pyramid with fast unit tests around the business core
Auditability	Signed points transactions, related order identifiers, timestamps, role checks, and explicit domain/application boundaries

The main trade-off is that the prototype favors local reproducibility over production scale: H2 and embedded Artemis reduce setup cost, while a production deployment would normally use a managed relational database, an external broker, CSRF protection, and external payment providers.

---

21. Transparency and Traceability

The traceability chain is:

```
Assignment requirement
    -> use case / quality attribute
    -> UML or GUI artifact
    -> controller/service/mapper implementation
    -> automated test
    -> known gap or completion status
```

The following archived statistics come from the supplied 7.1–7.3 workbooks and describe the commit that added those workbooks. They are retained as submission evidence rather than regenerated from the current working tree.

7.1 Package and class counts

Package	Description	Classes
`com.hotel.hotel_loyalty`	Spring Boot entry point	1
`.common`	Shared utilities and exception handling	3
`.config`	Security and JMS configuration	2
`.controller`	REST and JSP controllers	10
`.domain`	Database table data objects	6
`.dto`	Inbound request payloads	6
`.enums`	Type-safe enumerations	6
`.mapper`	MyBatis data access mappers	6
`.messaging`	Events and JMS messages	5
`.order`	Order state machine	6
`.payment`	Payment strategy pattern	8
`.security`	Login and authorization handlers	11
`.service`	Business service interfaces	4
`.service.impl`	Business service implementations	4
`.vo`	Outbound response view objects	11
Total	**15 packages**	**89**

7.2 Class authorship and lines of code

Package	Class	Author	LOC
com.hotel.hotel_loyalty	HotelLoyaltyApplication	wukewei	23
common	DateTimeDisplayFormatter	Fordring	23
common	GlobalExceptionHandler	wukewei	78
common	Result	wukewei	41
config	JmsConfig	wukewei	51
config	SecurityConfig	wukewei	112
controller	AdminPointsController	wukewei	30
controller	AdminPointsRuleController	wukewei	31
controller	AdminProductController	wukewei	40
controller	AuthJspController	wukewei	34
controller	DemoJspController	wukewei	36
controller	GuestJspController	Fordring	155
controller	GuestPointsController	Fordring	103
controller	PointsAdjustmentController	wukewei	53
controller	PointsRuleController	wukewei	70
controller	ProductManagementController	wukewei	83
domain	OrderDO	wukewei	60
domain	PaymentMethodDO	wukewei	42
domain	PointsRuleDO	wukewei	60
domain	PointsTransactionDO	wukewei	51
domain	ProductDO	wukewei	48
domain	UserDO	wukewei	53
dto	PointsAdjustDTO	wukewei	30
dto	PointsAdjustmentDTO	wukewei	28
dto	PointsRuleDTO	wukewei	48
dto	ProductCreateDTO	wukewei	41
dto	ProductUpdateDTO	wukewei	54
dto	RedemptionRequest	Fordring	44
enums	OrderStatusEnum	wukewei	59
enums	PaymentTypeEnum	wukewei	98
enums	PointsRuleTypeEnum	wukewei	57
enums	UserLevelEnum	wukewei	60
enums	UserRoleEnum	wukewei	51
enums	UserStatusEnum	wukewei	54
mapper	OrderMapper	wukewei	52
mapper	PaymentMethodMapper	wukewei	27
mapper	PointsRuleMapper	wukewei	46
mapper	PointsTransactionMapper	wukewei	34
mapper	ProductMapper	wukewei	60
mapper	UserMapper	wukewei	54
messaging	PointsAdjustmentConsumer	wukewei	76
messaging	PointsAdjustmentMessage	wukewei	43
messaging	PointsAdjustmentProducer	wukewei	38
messaging	PointsDeductedEvent	wukewei	43
messaging	RewardDistributionListener	wukewei	36
order	CancelledState	wukewei	37
order	CompletedState	wukewei	37
order	OrderContext	wukewei	76
order	OrderState	wukewei	32
order	PaidState	wukewei	39

Package	Class	Author	LOC
order	PendingPaymentState	wukewei	37
payment	PaymentAmountCalculator	Fordring	24
payment	PaymentRequest	wukewei	39
payment	PaymentResult	wukewei	31
payment	PaymentStrategy	wukewei	22
payment	PaymentStrategyFactory	wukewei	49
payment	PointsOnlyStrategy	wukewei	37
payment	PointsPlusCashStrategy	wukewei	48
payment	PointsPlusVoucherStrategy	wukewei	49
security	AppUserDetailsService	wukewei	92
security	LoginContext	wukewei	36
security	LoginFailureHandler	wukewei	39
security	LoginLogHandler	wukewei	26
security	LoginPostHandler	wukewei	55
security	LoginSuccessHandler	wukewei	70
security	PortalAwareAuthenticationEntryPoint	wukewei	37
security	PortalAwareLogoutSuccessHandler	wukewei	36
security	PortalMatchHandler	wukewei	27
security	RedirectHandler	wukewei	23
security	StatusCheckHandler	wukewei	23
service	GuestPointsService	Fordring	60
service	PointsAdjustmentService	wukewei	38
service	PointsRuleService	wukewei	39
service	ProductManagementService	wukewei	47
service.impl	GuestPointsServiceImpl	Fordring	329
service.impl	PointsAdjustmentServiceImpl	wukewei	86
service.impl	PointsRuleServiceImpl	wukewei	133
service.impl	ProductManagementServiceImpl	wukewei	162
vo	AccountVO	wukewei	47
vo	ConfirmPaymentVO	wukewei	36
vo	MyPointsVO	wukewei	39
vo	PaymentMethodVO	wukewei	19
vo	PointsHistoryVO	wukewei	36
vo	PointsMallVO	wukewei	21
vo	ProductAdminVO	wukewei	36
vo	ProductVO	wukewei	25
vo	RedemptionRecordVO	wukewei	49
vo	RuleAdminVO	wukewei	43
vo	UserPointsVO	wukewei	29

7.3 Team contribution

Team member	Classes	LOC	Share
wukewei	82	3,966	84.4369%
Fordring	7	731	15.5631%
Total	**89**	**4,697**	**100%**

The endpoint-to-use-case mapping remains in Section 8, and the test-to-behavior summary is in Section 12. These archived counts should not be presented as a fresh count if the source tree changes after the documented commit.

22. Critique of Design and Implementation

The design is mostly solid, but several features remain unfinished. Controllers do not access mappers directly, services do not call each other, and the layered MVC structure keeps responsibilities clear. The payment strategies and login validation chain are small and independently testable. Optimistic locking prevents concurrent balance or stock updates from overwriting one another. Unit tests cover the strategies and chain, while integration tests exercise the message flow with Artemis and H2; CI runs the automated tests on every push.

The main weaknesses are in the incomplete event-driven and operational features. There is no dead-letter queue, so messages that exhaust all retries are not administratively recoverable. The reward-distribution listener deducts stock inside the redemption transaction, but no real external delivery (e-mail confirmation, shipment, or voucher issuance) or guest notification is implemented yet. The queue architecture is prepared for batch issuing, but no batch-issuing endpoint exists and the 10,000-user scenario has not been tested. These limits are acceptable for an academic prototype but must be addressed for production use.

Utility of diagrams

The UML diagrams are useful implementation guides: the class, layered-package, and pattern diagrams closely match the package structure, and the use-case diagrams helped derive controller endpoints and test scenarios. Their limitation is that they are static and cannot be automatically checked against source code. For example, the order-state diagram describes a `refundPoints()` operation that is not present in the implementation. The diagrams should therefore document the main flows, omit details that change frequently, and be reviewed whenever the code changes.

---

23. Reflection

Fordring

This project taught me how Chain of Responsibility, Observer, Strategy, and State patterns are used in a real application rather than only in isolated examples. I also studied Spring Boot MVC, layered architecture, security, persistence, and asynchronous processing in more depth. During requirements confirmation, discussion of the key use cases revealed ambiguity and caused part of the work to be restarted, which affected the schedule. The team recovered by prioritizing the clarified use cases, working through the implementation and test evidence, and bringing the schedule back on track. The main lesson is that early use-case agreement is not administrative overhead: it is a technical dependency for architecture, implementation, and testing.

Franklin

What I learned

The most important thing I learned is the value of setting rules first. At the start, we wrote the architecture document, the coding standards (based on the Alibaba Java Development Manual), and the test strategy. When problems came up, we went back to these rules, found the cause, and improved the rules.

For example, the first version of selectAll() did not filter out items with no stock. We missed this while coding. Before writing the tests, we checked the code against the UC3 sequence diagram and found it. So the rules helped us find a mistake that the code alone did not show.

Project value

We worked in a “rules first” way: documents and standards came before the code, and the test strategy came before the tests.

What worked

Setting rules first for the “how” part, such as UML and the test strategy, gave us a clear base for later work.

What did not work

We drew the GUI prototypes for the “what” part too late, after the design phase. As a result, the team understood some requirements differently, such as what the “threshold” in the points rule meant. The lesson is to draw prototypes before the design, so everyone agrees on what we are building before we decide how.

---

24. Generative AI Use

The team used Copilot, DeepSeek V4 Flash, Qwen3-Coder 30B, and JoyAI Code 1.6 as engineering assistants. Human team members retained responsibility for requirements, verification, source-code decisions, and the final report.

#	Task	Representative prompt	AI output	Human action
1	CI/CD setup	“This GitHub assignment project has no server. How can we add CI/CD?”	`ci.yml` for push-to-Maven verification and `release.yml` for tag-to-JAR/GitHub Release	Chose branch and trigger rules; fixed a token-scope failure and a test failure; changed the test assertion
2	UML audit	“Check which UML diagrams are valid and match the code.”	Audited 15 diagrams; 9 matched and 6 had discrepancies such as incorrect class names, outdated claims, or incorrect SQL	Verified each finding against source code before accepting it
3	Test code	“Fix the failing PointsAdjustmentMessagingTest.”	Revised the test to assert that stock decreases by one, matching the listener-deducts-stock event-driven design	Confirmed the design intent and changed the test rather than the implementation
4	Concept explanations	“What does git tag mean?” and “How does GitHub Actions CI work?”	Explained Git tags and GitHub Actions CI	Used the explanations to improve understanding, then executed commands manually

Secrets, credentials, and unnecessary private data must not be included in the prompt history.

---

25. References and Submission Checklist

Formal reference list

1. Bass, Len, Paul Clements, and Rick Kazman (2013). *Software Architecture in Practice* (3rd ed.). Addison-Wesley. Used for architectural drivers, quality attributes, tactics, and architecture evaluation.

2. Gamma, Erich, Richard Helm, Ralph Johnson, and John Vlissides (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley. Used for Strategy, State, Observer, Factory, and Chain of Responsibility patterns.

3. Fowler, Martin (2002). *Patterns of Enterprise Application Architecture*. Addison-Wesley. Used for MVC, service-layer separation, Data Mapper, and enterprise application structuring.

4. Walls, Craig (2022). *Spring in Action* (6th ed.). Manning Publications. Used for Spring Boot, Spring MVC, dependency injection, web security, and application configuration.

5. Martin, Robert C. (2002). *Agile Software Development: Principles, Patterns, and Practices*. Prentice Hall. Used for the lifecycle discussion, iterative refinement, SOLID principles, and the balance between complexity management and agility.

The repository files, UML diagrams, use-case workbooks, and GitHub history are project evidence rather than bibliographic references. They are cited in the traceability matrix and supporting-artifact sections above.

Final export checklist

- [ ] Confirm the two page-limited discussions meet their page limits.
- [ ] Insert the two detailed key-use-case descriptions and the required diagrams.
- [ ] Refresh package/class/LOC/authorship tables from the final commit.
- [ ] Add the required version-control screenshots, if the marking format requires images.
- [ ] Replace the Franklin reflection placeholder with the team member's own account.
- [ ] Append the final AI prompt list and verify that it contains no secrets.
- [ ] Run `./mvnw clean test` and record the final test result and test count.
- [ ] Review all “gap” labels so planned work is not presented as implemented behavior.
- [ ] Export this README to PDF/HTML with SVG figures rendered and links preserved.

README Update Supplement

The original report pages are preserved. This page records the README changes incorporated into the PDF without regenerating the report.

Report metadata

Last reviewed: 4 November 2026

Section 13 - UML and Supporting Documents

The supporting-artifact table also includes the following entries:

Artifact	Location
Runtime component architecture	UML/component_diagram.svg
Single-host deployment topology	UML/deployment_diagram.svg

These entries correspond to the updated README section 13 and the existing SVG files.